TECHNICAL DOCUMENTATION

Online Retail Customer Segmentation using RFM Analysis and Clustering

1. Introduction

In today's retail industry, businesses generate a huge amount of transaction data every day. Understanding customer purchasing behavior is important because it helps companies improve customer satisfaction, increase sales, and build effective marketing strategies. Instead of treating every customer in the same way, businesses can segment customers into different groups and provide services based on their purchasing patterns.

In this project, I worked on an Online Retail dataset of a UK-based company that sells unique gift items. The main aim of this project was to analyze customer purchasing behavior and identify customer segments using RFM Analysis and clustering techniques.

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2. Problem Statement

The company has a large customer base with different purchasing habits and spending patterns. Identifying valuable customers manually is difficult and time-consuming. Therefore, customer segmentation is required to group customers based on their behavior and help businesses make better marketing decisions.

The goal of this project was to identify major customer segments and generate meaningful business insights using transaction data.

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3. Dataset Description

The dataset contains transaction records of a UK-based online retail company. It includes details such as invoice numbers, product descriptions, quantity purchased, unit price, customer information, and country details.

The dataset consists of:

- 541,909 records
- 8 features
- Customer transaction information collected between December 2010 and December 2011

Important columns used in this project were:

- CustomerID

- InvoiceNo
- Quantity
- UnitPrice
- InvoiceDate
- Country

These columns were mainly used for customer behavior analysis and RFM calculation.

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4. Data Understanding and Cleaning

Before starting the analysis, I explored the dataset using functions such as head(), tail(), shape(), info(), and describe() to understand its structure and quality.

While exploring the data, I found duplicate records and missing values. The CustomerID column contained a large number of missing values, and since customer identification is important for segmentation, rows containing missing CustomerID values were removed.

I also removed duplicate records and filtered out transactions with negative Quantity and UnitPrice values to ensure that only valid purchases were included in the analysis.

After completing these steps, the dataset became cleaner and more suitable for further analysis.

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5. Exploratory Data Analysis

After cleaning the data, I performed Exploratory Data Analysis to understand customer purchasing behavior and business trends.

Some important findings from the analysis were:

- The majority of transactions were generated from the United Kingdom.
- November recorded the highest number of transactions, followed by October and December.
- Customer activity was lower during February and April.
- Most purchases occurred between 12 PM and 3 PM.
- Afternoon was the most active time period for customer purchases.

I also analyzed top-selling products, stock codes, quantity distributions, and customer activity across different time periods. These visualizations helped me better understand customer behavior before performing segmentation.

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6. Feature Engineering

To gain additional insights, I created new features from the InvoiceDate column.

The new features included:

- Month
- Day
- Hour

I also created a new feature called TotalAmount by multiplying Quantity and UnitPrice.

This feature helped calculate the total spending of each customer and was later used in the RFM model.

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7. RFM Analysis

RFM Analysis is one of the most widely used techniques for customer segmentation.

RFM stands for:

- Recency – How recently a customer made a purchase.
- Frequency – How often a customer makes purchases.
- Monetary – How much money a customer spends.

Using CustomerID, I calculated Recency, Frequency, and Monetary values for every customer. These metrics helped measure customer value and purchasing behavior.

The resulting RFM table formed the foundation for customer segmentation.

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8. Data Transformation

The RFM variables were highly skewed, which could affect clustering performance.

To reduce skewness, I applied logarithmic transformation to the RFM variables. After transformation, the distributions became more balanced.

I then applied StandardScaler to standardize the features because clustering algorithms perform better when all variables are on a similar scale.

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9. Customer Segmentation using Clustering

After preparing the RFM dataset, I applied clustering algorithms to identify customer groups.

Initially, I used K-Means Clustering and evaluated different values of K using the Elbow Method and Silhouette Score.

I also performed Hierarchical Clustering and analyzed the dendrogram to validate the results.

Based on the clustering evaluation techniques used in the project, the optimal number of customer segments was identified as two.

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10. Customer Segment Analysis

The final clustering results divided customers into two groups.

Cluster 0

Customers in this segment showed relatively lower spending and lower purchase frequency compared to the second cluster.

Business Recommendation:

- Provide promotional offers.
- Run retention campaigns.
- Encourage repeat purchases through discounts and rewards.

Cluster 1

Customers in this segment showed higher spending and higher purchase frequency. These customers contribute significantly to business revenue.

Business Recommendation:

- Loyalty programs.
- Personalized recommendations.
- Exclusive offers and premium customer benefits.

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11. Key Business Insights

The project generated several useful business insights:

- Most customers belong to the United Kingdom.
- Sales activity increases significantly during October, November, and December.
- Customers are most active during afternoon hours.
- Customer behavior can be effectively segmented using RFM Analysis.
- Two distinct customer groups were identified based on purchasing behavior.

These findings can help businesses improve customer engagement and design targeted marketing strategies.

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12. Conclusion

This project successfully analyzed customer transaction data and identified meaningful customer segments using RFM Analysis and clustering techniques.

The project involved data cleaning, exploratory data analysis, feature engineering, customer behavior analysis, and clustering. Using K-Means Clustering and Hierarchical Clustering, customers were segmented into two groups based on their purchasing behavior.

The insights generated from this analysis can help businesses identify valuable customers, improve customer retention, create personalized marketing campaigns, and make better business decisions. This project also helped me gain practical experience in data analysis, customer analytics, feature engineering, and unsupervised machine learning techniques.